

NVH Source Locator — ■■■■■■



- ■■■
- ■■■■■
- ■■■■■
- 2-Sensor■■■
- 3-Sensor■■■
- Pro+■■■ (3-Sen+, 4-Sensor, 4-Sen+, 3D, 3D+)
- Materials■■
- ■■■■
- ■■■■■
- ■■■■
- ■■■■■■■■■■
- ■■
- Pro■■
- Help■■■■■■■■■■
- ■■■■■■■■■■

■ ■ ■ {#how-it-works}

2

NVH Source Locator■■■■■■■■■■:

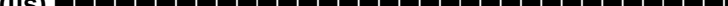
- [illegible]

[illegible][illegible]

- 2  $\rightarrow$ 
- 3  $\rightarrow$ 2D  (X, Y)
- 4  $\rightarrow$ 3D  (X, Y, Z)

■ ■ ■ ■ ■ {#before-you-start}

□□□□□:

-  (μs)
-  2  ()
-  ()
-  ()

**NVH Source Locator**

TDOA Calculator

PRO



2-Sensor

3-Sensor

3-Sen+

4-Sensor

4-Sen+

3D

3D+

Materials

Pair A-B

Pair A-C

Step 1

Calibration — Speed of Sound

Sensor spacing A-B

-

118

+

cm

External knock time delay

-

471

+

μs

Calculated speed of sound

2505 m/s

Closest: PEEK (2500 m/s)

+ Save as custom material

Step 2

Event Measurement

Event time delay

-

227

+

μs

⊕ trials

Which sensor heard it first?

Sensor A

Calculate Source Location

2-Sensor

■■■■■ {#the-main-tabs}

■■■■■■■■■■■.

2-Sensor	21D	<>
3-Sensor	32D	
3-Sen+3-Sensor	3-Sensor	
4-Sensor	22D (A-B + C-D)	
4-Sen+4-Sensor	22D4LSQ	
3D3D	XYZ43D3D	
3D+3D	63DLSQ	
Materials	+	1
Help		
 vs Pro: 2-SensorPro Pro paywall		

[illegible]



NVH Source Locator

TDOA Calculator PRO




2-Sensor
3-Sensor
3-Sen+
4-Sensor
4-Sen+
3D
3D+
Materials

Pair A-B

Pair A-C

Step 1

Calibration – Speed of Sound

Sensor spacing A-B

–

118

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cm

External knock time delay

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471

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μs

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2505 m/s

Closest: PEEK (2500 m/s)

+ Save as custom material

Step 2

Event Measurement

Event time delay

⊕ trials

–

227

+

μs

Which sensor heard it first?

Sensor A

Calculate Source Location

2-Sensor

1:

Materials : Mild
(1020)

2

2:

2-Sensor **2** **A-B** **A-C**

2 **A-B**

:

- **(d):** **cm**
- **(tCal):** —

3:

- **(tEvent):**
- **:** **(A**

4:

A

- **= 0:** **A**
- **= :** **B**
- **:**
- **:** **toast**

A **B**

5:

A **B**

3-Sensor **{#3-sensor-mode}**

3 **2D**

NVH

NVH Source Locator

TDOA Calculator

PRO

2-Sensor

3-Sensor

3-Sen+

4-Sensor

4-Sen+

3D

3D+

Materials

Same triangle as 3-Sensor, but calibrate & measure **all three pairs** (A-B, A-C, B-C). The solver uses all 3 TDOAs in a least-squares fit – more robust to measurement noise and anisotropic materials. The *RMS residual* tells you how consistent your measurements are.

▲ Show placement guide

Setup

Triangle side lengths

A-B side length

-

100

+

cm

A-C side length

-

90

+

cm

B-C side length

-

80

+

cm

Pair A-B

Calibration & measurement

Calibration time delay – knock outside A-B

-

400

+

μs

2,500 m/s – Closest: PEEK (2,500 m/s)

Event time delay

-

120

+

μs

⊕ trials

Which sensor heard it first?

Sensor A

3-Sensor

3

(A-B A-C B-C)

(A-B A-C):

3-Sensor (A-B, A-C, B-C)
3 TDOA —

4-Sensor

4:

- A-B = /
- C-D = /

A-B C-D 2D —

4-Sen+ (2D)

4 A B C D

3D

3D 4 3D (X, Y, Z) (A-B, A-C, A-D)

3D+ (Pro)

3D LSQ 6 (A-F) 3D

Materials {#the-materials-tab}

20 °C

NVH

NVH Source Locator

TDOA Calculator

PRO

2-Sensor3-Sensor3-Sen+4-Sensor4-Sen+3D3D+Materials

Typical longitudinal wave speeds at ~20 °C (sources: Evident/Olympus NDT & Dakota Ultrasonics). Actual speeds vary with composition, microstructure, temperature, and grain orientation – always calibrate in-situ for best accuracy. **Tap a material to apply its speed across every tab** – calibration Δt is computed from each pair's distance. Pairs without a distance entered are skipped.

Q

Search...

Air

343 m/s

ref only

☆

Teflon (PTFE)

1,400 m/s

ref only

☆

Mercury

1,450 m/s

ref only

☆

Water

1,480 m/s

ref only

☆

Silicone rubber

1,485 m/s

ref only

☆

Neoprene

1,600 m/s

ref only

☆

Motor Oil (SAE 30)

1,740 m/s

ref only

☆

Rubber, Butyl

1,800 m/s

ref only

☆

Polyurethane

1,900 m/s

ref only

☆

Glycerine

1,920 m/s

ref only

☆

LDPE

2,080 m/s

ref only

☆

Lead

2,160 m/s

ref only

☆

NVH

NVH Source Locator

TDOA Calculator

PRO

2-Sensor3-Sensor3-Sen+4-Sensor4-Sen+3D3D+Materials

Platinum

3,300 m/s

ref only

☆

Tin

3,320 m/s

ref only

☆

Uranium

3,378 m/s

ref only

☆

Bronze

3,530 m/s

☆

Silver

3,607 m/s

ref only

☆

Ice

4,000 m/s

ref only

☆

Concrete

4,000 m/s

ref only

☆

Zinc

4,170 m/s

☆

Brass

4,430 m/s

☆

Iron (cast)

4,572 m/s

☆

Zirconium

4,650 m/s

ref only

☆

Copper

4,660 m/s

☆

Tungsten

5,180 m/s

☆

Glass (crown)

5,300 m/s

ref only

☆

Monel

5,359 m/s

ref only

☆

Nickel

5,630 m/s

☆

Quartz

5,740 m/s

ref only

☆

Materials

~340 m/s ~13,000 m/s

14~20 °C:

-
- Mild (1020)
- (304)
- ()
-
-
-
-
-
-

- 2
-
-
-
-

2: 20 °C
"ref only" —

2-Sensor 2-sensor

in-situ

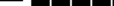
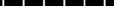
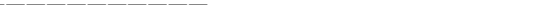
{#temperature-compensation}

NVH: 80 °C -10 °C 200 °C

() → °C -40 +200

 Save image



-   — 
 - 
 - 
 -  A  B  C  D  E  F  —
-  6 

- [REDACTED]
 - [REDACTED]
- [REDACTED]PDF[REDACTED]

[REDACTED] {#reports}

[REDACTED]



3-Sensor Triangle Source Location Result
NVH Source Locator – Report

MEASUREMENT INPUTS

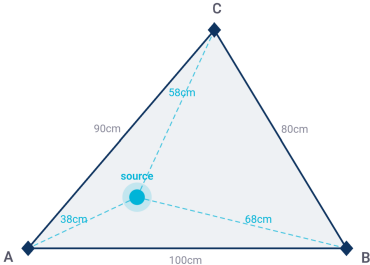
A–B side length	100cm
A–C side length	90cm
B–C side length	80cm
A–B calibration Δt	400μs
A–B event Δt	120μs
A–C calibration Δt	360μs
A–C event Δt	80μs
Distance from A	37.7cm
Distance from B	67.7cm
Distance from C	57.7cm
Fit residual	0.00

CONCLUSION

Source is inside the triangle – closest to sensor A

A, B, C = sensor corner positions

VISUALISATION
3-Sensor Triangle Source Location Result



PDF[REDACTED]

[REDACTED]

- [REDACTED] → [REDACTED]
- [REDACTED]
- [REDACTED]
- [REDACTED]
- [REDACTED]
- [REDACTED]
- [REDACTED]
- [REDACTED]
- [REDACTED]
- [REDACTED]

[REDACTED]

- Android: [REDACTED]PDF[REDACTED]
- iOS: [REDACTED] → PDF[REDACTED]AirPrint[REDACTED]

Units & settings

×

LANGUAGE

English

DISTANCE

cm

inches

TIME

μs

ms

REFERENCE TEMPERATURE

Resets to 20 °C on next app launch

-

20

+

°C

Used for temperature-compensated material speeds. Always calibrate in-situ for best accuracy.

MEASUREMENT HISTORY

🕒 View recent calculations

⬇️ Backup

⬆️ Restore

REPORT

Report header text

e.g. EVDiag NVH Services · service@evdiag.net · +49-XXX-XXXXX

Appears at the top of every printed report.

0 / 500

DATA

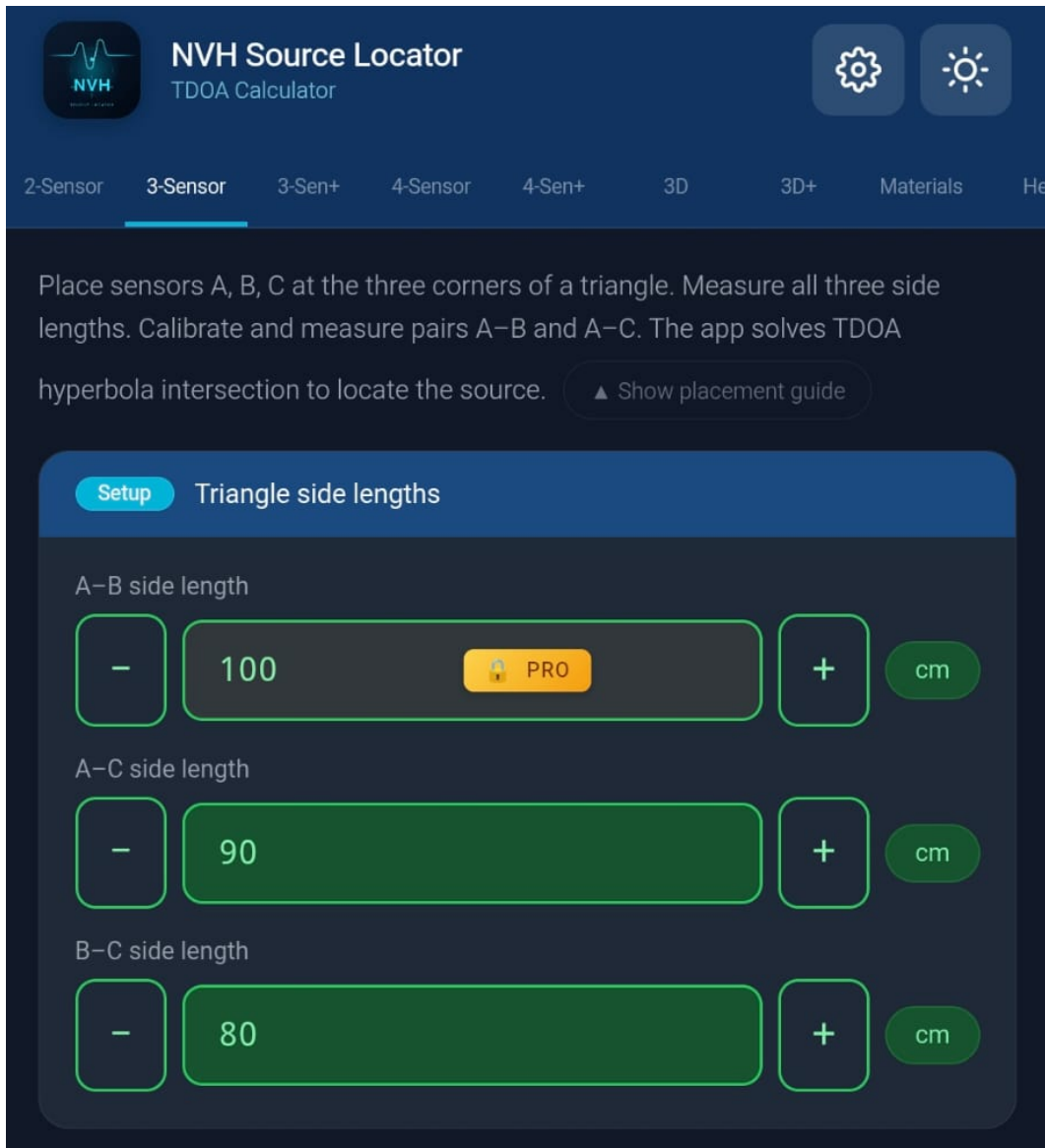
🕒 Show intro screen again

ABOUT

NVH Source Locator · TDOA Calculator for accelerometer-based source localization. Works offline. No data leaves your device.



Pro 	Pro (\$19.99)
 	30
 	
 	cm
 	-40 +200 °C
 	



Paywall



Questions? support@evdiag.net

Paywall

- [illegible]

Pro

Pro
AndroidGoogle PlayiOSApple App
Store

Pro

- 1Pro:
- GoogleAndroidApple IDiOS
 - NVH Source Locator
 -
 - Pro

NVH Source LocatorGoogle Play StoreApp
StorePro —

Android: paypalGoogle PlayGoogle
Play

iOS: App Store 3.1.1 AppleGoogle
PlayiOSApp Store

Help {#help-tab-and-tutorials}

Help



Works offline

Installable on Android & iOS

No data sent anywhere

Single pair A-B or A-C. Linear result bar.



Two pairs, proportional rectangle map.



Triangle mode, TDOA hyperbola solver.



49 materials. Tap to auto-fill calibration.



Mode explainers, placement guides, and a 10-item FAQ covering calibration, TDOA theory, and accuracy limits.

1

Strike the component outside the sensor pair. Enter the sensor spacing (cm) and the measured time delay (μs) – the app calculates the speed of sound in the material.

Help

□ □ □ □ □ □ □ □ □ □ :

- 
- 
- 
- 
- 

- ■■■■■■■■■■support@evdiag.net■■■■■■■■

0

0
-40/+200

■■■■■■■■■■

support@evdiag.net ■■■■■■■■■■:

-  OS 
-  → 
- 
- 

[illegible]